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CausalTables.jl: Simulating and storing data for statistical causal inference in Julia

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Summary

Estimating the strength of causal relationships between variables is an important problem across many scientific disciplines. `CausalTables.jl` provides tools to (1) easily store and process tabular data endowed with causal structure, (2) simulate data from causal models for experimental testing, and (3) compute ground-truth estimates of causal parameters. Together, these functionalities expand the Julia ecosystem by supporting the development and experimental assessment of new statistical causal inference methods in Julia.

Statement of need

The quantitative science of causal inference has emerged over the past three decades as a set of formalisms for studying cause-and-effect relationships between variables from observed data (Hernán & Robins, 2020; Pearl, 2009). Causal inference techniques have helped applied scientists and decision-makers better understand important phenomena in fields ranging from health and medicine to politics and economics. New software tools for causal inference are being developed at a rapid pace, but in the Julia language, there currently do not exist auxiliary tools designed to support their development. `CausalTables.jl` aims to provide such a tool.

Attempts to implement and test causal inference methods in Julia face two major challenges. First, statistical causal inference requires data to be transformed in various ways based on the underlying causal structure, in order to be provided as input to other packages. For instance, causal methods using `MLJ.jl` (Blaom et al., 2020) would necessitate extracting the variables hypothesized to cause the treatment and response as Tables; meanwhile; using `GLM.jl` (Bates et al., 2023) would require the same but as Matrix or `Formula` objects. Second, testing the performance of new estimators often requires simulating data for numerical experiments from a Structural Causal Model (SCM) (Pearl, 2009) so as to compare them to an underlying ground truth (encoded via interventions on the SCM).

`CausalTables.jl` provides an interface to solve these two problems, simplifying the development of packages for statistical causal inference on tabular data in Julia. It implements a `CausalTable` interface that extends `Tables.jl`, the most common interface for accessing tabular data in Julia (Quinn et al., 2024). The package also implements a `StructuralCausalModel` interface for sampling from any SCM and computing ground-truth estimates of causal parameters. This interface operates in conjunction with `Distributions.jl`, the primary Julia package for working with random variables (Besançon et al., 2021; Lin et al., 2019). By integrating seamlessly with other commonly used packages in the Julia ecosystem, `CausalTables.jl` ensures both compatibility and ease of use for statisticians and applied scientists alike.

Comparison to existing packages

While the R and Python ecoysystems include many implementations of causal methods (Chen et al., 2020; van der Laan et al., 2024), Julia has relatively fewer. Recent Julia packages for causal inference include `TMLE.jl` (Labayle et al., 2024) and `CausalELM.jl` (Colby, 2024). These packages focus on specific estimation techniques, each implementing different interfaces to label causal structure for their specific problems. They do not provide a general simulation or causal-specific data processing framework like `CausalTables.jl`. On the other hand, `CausalInference.jl` (Schauer et al., 2024) provides an interface for representing causal graphs and implements causal discovery algorithms, similar to CausalDAG (Squires, 2018) or DoWhy (Sharma & Kiciman, 2020) in Python and daggity (Textor et al., 2017) in R. However, it is generally incompatible with the tabular data format required by statistical tools, and also cannot simulate data. In fact, as far as we are aware, `CausalTables.jl` is the first package for simulating and extracting ground-truth causal estimands from an existing SCM in Julia.

Example 1: Data Preprocessing

`CausalTables.jl` supports causal inference problems that involve estimating the effect of at least one treatment variable A on a response variable Y. Using the `CausalTable` constructor, one can wrap existing data as a `Tables.jl`-compliant structure coupled with causal structure labels.

```julia
using CausalTables

# Example data in Tables-compatible format
tbl = (W = [0.2, 0.4, 0.7],
       A = [false, true, true],
       Y = [0.8, 1.2, 2.3])

# Wrap data as CausalTable
ct_wrap = CausalTable(tbl; treatment = :A, response = :Y, confounders = [:W])
```

Convenience functions perform data processing tasks common to causal inference, such as selecting or intervening on specific variables. For example, the `parents` function can be used to select only variables denoted as causes of Y:

```julia
parents(ct_wrap, :Y)
```

```
CausalTable
┌────────┬───────┐
│      W │     A │
│ Float64 │  Bool │
├────────┼───────┤
│    0.2 │ false │
│    0.4 │  true │
│    0.7 │  true │
└────────┴───────┘
Summaries: NamedTuple()
Arrays: NamedTuple()
```

Example 2: Simulating data with ground-truth approximations

An SCM defines causal structure by envisaging a data-generating process as random draws from a sequence of non-parametric structural equations, with each draw depending on realizations from draws preceding it. An example is the following:

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$$
\begin{aligned}
W &\sim Beta(2, 4) \\
A &\sim Bernoulli(W) \\
Y &\sim Normal(A + W, 1)
\end{aligned}
$$

Using `CausalTables.jl`, we can define the SCM above by enumerating the sequence of random variables to be drawn, along with labels for the role they play in estimation, and then randomly draw from it.

```julia
using Distributions

# Define the sequence of random variables to be drawn
dgp = @dgp(
    W ~ Beta(2, 4),
    A ~ Bernoulli.(0.5 .* W .+ 0.2),
    Y ~ Normal.(W .+ A, 1)
)

# Create a structural causal model (SCM) from the DGP
scm = StructuralCausalModel(dgp;
  treatment = :A, response = :Y, confounders = [:W]
)

ct = rand(scm, 500) # randomly draw from the SCM
```

`CausalTables.jl` provides high-level functions that approximate ground truth values of common causal estimands when called on the `scm`. These include:

- Average treatment effects (`ate`) including among the treatment (`att`) and untreated (`atu`)
- Counterfactual means (`cfmean`) and differences (`cfdiff`)
- Average policy effects (`ape`)

In addition, `CausalTables.jl` implements low-level interface for (1) applying common interventions to the treatment variable in a `CausalTable`, (2) drawing randomly from counterfactual distributions, and (3) computing ground truth conditional densities and functions of these (e.g., means, variances, propensity scores), which often arise in the definition of many estimands.. For example, below we compute the difference in the conditional mean of Y under treatment versus no treatment, the difference of which is the ATE.

```julia
treated = intervene(ct, treat_all)    # CausalTable with everyone treated
untreated = intervene(ct, treat_none) # CausalTable with no one treated
mean(conmean(scm, treated, :Y) .- conmean(scm, untreated, :Y))
```

```
1.0
```

Closing remarks

`CausalTables.jl` provides useful auxiliary functions to support causal inference methods on tabular data in Julia that involve one or more treatment variables and responses. Users can simulate data from any SCM and benchmark methods using either high-level functions for common estimands or low-level functions for more exotic estimands. By combining this with the power of the `CausalTable` interface for processing data once it is generated, `CausalTables.jl` serves as a useful tool for scientists seeking to develop and experimentally evaluate new causal inference methods.

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References

Bates, D., Noack, A., Kornblith, S., Bouchet-Valat, M., Borregaard, M. K., Arslan, A., White, J. M., Kleinschmidt, D., Alday, P., Lynch, G., Dunning, I., Mogensen, P. K., Lendle, S., Dilum Aluthge, Mousum Dutta, Pdeffebach, José Bayoán Santiago Calderón, P., Ayush Patnaik, Born, B., … König, B. (2023). *JuliaStats/GLM.jl: v1.9.0*. Zenodo. https://doi.org/10.5281/ZENODO.3376013

Besançon, M., Papamarkou, T., Anthoff, D., Arslan, A., Byrne, S., Lin, D., & Pearson, J. (2021). Distributions.jl: Definition and modeling of probability distributions in the JuliaStats ecosystem. *Journal of Statistical Software*, *98*(16), 1–30. https://doi.org/10.18637/jss.v098.i16

Blaom, A. D., Kiraly, F., Lienart, T., Simillides, Y., Arenas, D., & Vollmer, S. J. (2020). MLJ: A julia package for composable machine learning. *Journal of Open Source Software*, *5*(55), 2704.

Chen, H., Harinen, T., Lee, J.-Y., Yung, M., & Zhao, Z. (2020). CausalML: Python package for causal machine learning. *arXiv Preprint arXiv:2002.11631*.

Colby, D. (2024). *CausalELM.jl*. https://github.com/dscolby/CausalELM.jl

Hernán, M. A., & Robins, J. M. (2020). *Causal inference*. CRC Boca Raton, FL.

Labayle, O., Beentjes, S., Khamseh, A., & Ponting, C. (2024). *TMLE.jl*. https://github.com/olivierlabayle/TMLE.jl

Lin, D., White, J. M., Byrne, S., Bates, D., Noack, A., Pearson, J., Arslan, A., Squire, K., Anthoff, D., Papamarkou, T., Besançon, M., Drugowitsch, J., Schauer, M., & contributors, other. (2019). *JuliaStats/Distributions.jl: a Julia package for probability distributions and associated functions*. https://doi.org/10.5281/zenodo.2647458

Pearl, J. (2009). *Causality*. Cambridge university press.

Quinn, J., Kamiński, B., Anthoff, D., Bouchet-Valat, M., Papp, T. K., Arakaki, T., Schouten, R., Robinson, N., mathieu17g, Samuel, O., Revels, J., ExpandingMan, Hanson, E., Blaom, A., Arslan, A., Ling, J., Chen, J., Day, J., Calderón, J. B. S., … Adenbaum, J. (2024). *JuliaData/Tables.jl: v1.12.0* (Version v1.12.0). Zenodo. https://doi.org/10.5281/zenodo.12753139

Schauer, M., Keller, M., & Wienöbst, M. (2024). *CausalInference.jl*. Zenodo. https://doi.org/10.5281/ZENODO.13684767

Sharma, A., & Kiciman, E. (2020). DoWhy: An end-to-end library for causal inference. *arXiv Preprint arXiv:2011.04216*.

Squires, C. (2018). *`causaldag`: creation, manipulation, and learning of causal models*. https://github.com/uhlerlab/causaldag

Textor, J., Zander, B. van der, Gilthorpe, M. S., Liśkiewicz, M., & Ellison, G. T. H. (2017). Robust causal inference using directed acyclic graphs: The r package "dagitty." *International Journal of Epidemiology*, dyw341. https://doi.org/10.1093/ije/dyw341

van der Laan, M., Coyle, J., Hejazi, N., Malenica, I., Phillips, R., & Hubbard, A. (2024). *Targeted Learning in R*. https://tlverse.org/tlverse-handbook/

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